

CY 644: Bioinorganic chemistry

Course Code	CY 644
Course Name	Bioinorganic chemistry
L-T-P-C	3-0-0-3
Prerequisites	CY 101 or equivalent and basic knowledge in coordination chemistry
Intended for	B. Tech. & Ph.d.
Distribution	Elective
Approval	4th Senate (May 2013)

Course Contents

- **Introduction** Fundamentals of coordination chemistry; Introduction to bioinorganic chemistry; Role of alkali and alkaline earth metal ions; Essential and trace metals; fundamentals of Na-K Pump; Ionophores and crown ethers. [7 Lectures]
- **Module II**
The role of metal ions in biological functions Their active structure/site and function in: (a) Metal ion transport and storage - Ferritin, Transferrin, metallothionein etc; (b) Electron Transfer - cytochromes, Fe-S Proteins, copper protein etc. (c) Oxygen activation - cytochrome P450, cytochrome c oxidase. (d) Oxygen transport and storage - hemoglobin, myoglobin, hemerythrin, hemocyanin, their coordination geometry and electronic structure, co-operativity effect, Hill coefficient and Bohr Effect; characterization of O₂ bound species by spectroscopic methods. [12 Lectures]
- **Module III**
Other enzymes - Peroxidase, alcohol dehydrogenase, carbonic anhydrase, nitrogenase, vitamin B₁₂ coenzyme. [2 Lectures]
- **Module IV**
The use of metals in medicinal chemistry- The significance of metal-based compounds for medicinal use (cis-platin, radiopharmaceuticals etc); recent developments in the medical field based on inorganic or bioinorganic principles; Metals used in diagnosis; Gd-based MRI and X-ray contrast agents. [8 Lectures]
- **Module V**
Some related basic tools and techniques - Bioassay, EPR spectroscopy, Single crystal X-ray diffraction, Electrochemistry with cyclic voltammetry, UV-Vis Absorption Spectroscopy, Fluorescence emission spectroscopy and Fluorescence excited state life time measurements. [5 Lectures]
- **Module VI**
Toxicity of metals Toxic effects of Pb, Cd, Hg and Cr with specific examples. [7 Lectures]
- **Module VII**
Metal Ions in Plant-Based Systems: The essential role of metal ions in photosynthesis and other processes in plant and fungal systems. Identify the bioinorganic molecules responsible for energy harvesting and electron transfer in biological systems. [4 Lectures]

Text Books

1. Stephen J. Lippard and Jeremy M. Berg, **Principles of Bioinorganic Chemistry**.

References

1. Wolfgang Kaimand, Brigitte Schwederski, **Bioinorganic Chemistry: Inorganic Elements in the Chemistry of Life**.
2. Ivano Bertini, Harry B. Gray, Edward I. Stiefel, Joan Selverstone Valentine, (Editors), **Biological Inorganic Chemistry**.
3. Wolfgang Kaimand, Brigitte Schwederski, **Bioinorganic Chemistry**.
4. Asim K. Das, **Bioinorganic Chemistry**
5. D. F. Shriver, P. W. Atkins, C. H. Langford, **Inorganic Chemistry**, Oxford University Press, 1990.
6. J. E. Huheey, E. A. Keiter, R. L. Keiter, **Inorganic Chemistry, Principles of Structure and Reactivity**, Pearson Education, 2004.